



Evolution of Machine Learning-Based Human-Disease Prediction Models: A Bibliometric Analysis of Statistical Learning, Model Validation, and Predictive Performance

Sarad Chandra Kafle ¹, Satyendra Kushwaha ², Basanta Prasad Adhikari ³, and Pradip Subedi ⁴

¹Asst. Prof., Birendra Multiple Campus, Tribhuvan University, Bharatpur, Chitwan, Nepal

²PhD Scholar, Faculty of Management, South Asian University, Delhi, India

³Prof., Department of Research, Oxford College of Engineering and Management, Nawalpur, Nepal

⁴IT Head, Birendra Multiple Campus, Tribhuvan University, Bharatpur, Chitwan, Nepal

Received: 10th June, 2026

Accepted: 12th August, 2026

Published: 15th September, 2026

ABSTRACT

Background: This study aims to provide a comprehensive bibliometric examination of machine learning applications in disease prediction, focusing on the evolution of machine learning frameworks and predictive performance from 2000 to July 2026 publication records. Using data from the Dimensions AI databases, this research maps the field's intellectual landscape by identifying the most influential authors, organisations, and emerging thematic trends.

Method: The study used bibliometric analysis and science mapping of records retrieved from Dimensions AI. The literature identification and screening process was reported using a PRISMA 2020 framework. The dataset was cleaned and analysed using Bibliometrix/Biblioshiny and VOSviewer, with performance analysis, collaboration analysis, title-term analysis, and thematic mapping.

Results: Publication growth accelerated sharply and reached its highest level in 2025, while the citation trend showed a notable peak in 2011. The study identified strong international collaboration, with China, the United States, India, and the United Kingdom among major contributors. Recent research has increasingly emphasized disease, patients, healthcare, deep learning, artificial intelligence, validation, and clinical prediction.

Conclusion: The research landscape has evolved from algorithm-oriented development toward clinically oriented, interpretable, and validated prediction models. Future research should prioritize disease-specific prediction using deep learning together with prospective validation, transparency, fairness, generalisability, and clinical implementation.

Keywords: Bibliometric Analysis, Clinical Validation, Deep Learning, Human-Disease Prediction, Machine Learning

Correspondence: Sarad Chandra Kafle, Birendra Multiple Campus, Tribhuvan University, Bharatpur, Chitwan, Nepal. E-mail: sarad.kafle49@gmail.com.

INTRODUCTION

The growing availability of electronic health records (EHRs), biomedical databases, genomic information, and high-dimensional clinical data has shifted modern healthcare research from conventional hypothesis-driven analysis to data-intensive predictive modeling (Wu et al., 2016). Early identification of individuals at risk of disease is a fundamental objective of precision medicine because accurate prediction models can support clinical decision-making, optimise resource allocation, and enable timely preventive interventions (Chowdhury and Turin, 2020). Clinical prediction models traditionally relied on statistical approaches such as logistic regression, Cox proportional hazards models, and survival analysis, where pre-defined relationships between predictors and outcomes were estimated using regression-based frameworks. These methods have provided a robust foundation for medical risk prediction because of their interpretability, statistical inference capabilities, and established methodological principles (Saelmans et al., 2025).

However, the rapid growth of healthcare data complexity has challenged traditional statistical approaches,

particularly when dealing with nonlinear relationships, large numbers of predictors, unstructured data, and complex interactions among clinical variables (Dhillon et al., 2022). Machine learning (ML), a subset of artificial intelligence (AI), has emerged as a powerful analytical paradigm that can identify hidden patterns in large-scale datasets without explicitly specifying underlying relationships (Scott, 2021). Machine learning algorithms, including random forests, support vector machines, gradient boosting methods, artificial neural networks, and deep learning architectures, have increasingly been applied for disease prediction across diverse clinical domains, including cardiovascular diseases, cancers, neurological disorders, infectious diseases, and metabolic conditions (Al-Nafjan et al., 2025).

The growing adoption of machine learning in disease prediction has generated substantial enthusiasm because of its potential to improve predictive accuracy and personalise healthcare delivery. Unlike conventional statistical models that often rely on linear assumptions and manually selected predictors, ML approaches can capture complex interactions, automate feature selection, and process heterogeneous data sources (Kingsmore et al.,

2021). Nevertheless, improved predictive performance alone does not guarantee clinical usefulness. A clinically applicable prediction model requires rigorous assessment of discrimination, calibration, robustness, generalisability, and external validity (Collins et al., 2015). Recent methodological research has highlighted that many ML-based clinical prediction studies suffer from inadequate validation strategies, insufficient reporting, overfitting risk, and limited assessment of model reliability (Andaur Navarro et al., 2023).

Although machine learning algorithms often show high predictive performance in development datasets, concerns remain about whether these improvements reflect genuine clinical advantages or are driven by methodological limitations (Andaur Navarro et al., 2021). A systematic review comparing machine learning approaches with logistic regression for clinical prediction models found no consistent evidence that ML methods universally outperform traditional statistical models, emphasising the importance of appropriate study design, transparent reporting, and rigorous evaluation procedures (Christodoulou et al., 2019). Similarly, investigations of ML-based prediction models have reported frequent deficiencies in sample size justification, missing-data handling, calibration assessment, and external validation, which can limit translation into real-world healthcare practice (Andaur Navarro et al., 2023).

Consequently, the evolution of machine learning-based disease prediction represents not only a technological advancement but also a significant methodological transition within statistical learning and predictive analytics (Zhang et al., 2022). Understanding how research priorities, analytical approaches, validation practices, and predictive evaluation metrics have developed over time is essential for identifying emerging trends and addressing current challenges. While several systematic reviews have evaluated specific diseases or particular ML applications, a comprehensive quantitative assessment of the global research landscape integrating statistical learning methods, model validation strategies, and predictive performance evaluation across disease prediction research remains limited (Andaur Navarro et al., 2023; Li et al., 2026).

Bibliometric analysis provides a systematic approach to examining the intellectual structure, evolution, collaboration patterns, influential publications, and emerging research themes within a scientific field (Donthu et al., 2021). Through techniques such as citation analysis, co-authorship analysis, keyword co-occurrence mapping, thematic evolution analysis, and science mapping, bibliometric studies can reveal how research domains develop and identify future directions (Aria and Cuccurullo, 2017). In the context of machine learning-based disease prediction, such an analysis can provide valuable insights into algorithmic trends, methodological transitions, geographic contributions, and the progression from traditional statistical modeling to advanced predictive systems (Li et al., 2026).

Therefore, this study aims to conduct a comprehensive bibliometric analysis of global research on machine learning-based disease prediction models. Specifically,

this study seeks to characterise publication growth and citation patterns, identify influential authors, institutions, countries, and journals contributing to this field, examine the evolution of machine learning and statistical learning approaches used for disease prediction, analyse trends related to model validation and predictive performance assessment, and identify emerging research themes and future directions in ML-driven disease prediction.

The study addresses the following research questions:

- RQ1:** What are the trends in publication growth and citation impact of machine learning-based human disease prediction research over time?
- RQ2:** Which authors, institutions, countries, and journals are most productive and influential, and what collaboration patterns characterize the field?
- RQ3:** How have machine learning approaches and research themes in disease prediction evolved over time?
- RQ4:** What emerging, motor, and declining themes shape the current and future trajectory of machine learning-driven disease prediction research?

LITERATURE REVIEW

Machine learning (ML) has evolved from an alternative computational approach to conventional statistical methods into a key framework for disease diagnosis, prognosis, and clinical risk prediction (Zhang et al., 2022). Early work by Kononenko (2001) established ML's potential for medical diagnosis, while advances in computational capacity and the availability of high-dimensional clinical data enabled the use of diverse algorithms, including decision trees, support vector machines, random forests, neural networks, and deep learning. Recent disease-specific reviews further show the expansion of ML across chronic, neurological, and infectious diseases, reflecting a shift from relatively simple predictive algorithms to increasingly complex, data-intensive approaches (Fatima and Masood, 2024; Islam et al., 2024; Qasrawi et al., 2025). This expansion, however, has also raised fundamental questions regarding whether increasing algorithmic complexity necessarily translates into superior clinical prediction.

Evidence comparing ML with conventional statistical learning provides a more nuanced perspective. A systematic review of 71 studies comparing ML with logistic regression found no evidence of superior predictive performance for ML when analyses were restricted to low-risk-of-bias studies. Importantly, 68% of studies had potential bias in validation procedures, while 79% did not assess calibration, suggesting that reported differences in predictive performance may partly reflect methodological weaknesses rather than genuine algorithmic superiority (Christodoulou et al., 2019). These findings highlight a broader transition in prediction research from an emphasis on algorithm selection and discrimination toward rigorous assessment of calibration, validation, generalisability, and clinical utility. Consistent with this development, Binuya et al. (2022) emphasised that evalua-

tion and updating of prediction models should consider discrimination, calibration, and clinical usefulness rather than relying on a single performance measure.

The growing methodological sophistication of ML prediction has also exposed persistent weaknesses in transparency and reproducibility. Andaur Navarro et al. (2023), reviewing 152 supervised ML-based prediction studies, found a median adherence of only 38.7% to TRIPOD reporting items, with particularly poor reporting of model specification, model-building procedures, participant flow, and predictive performance; external validation was also uncommon. These limitations have contributed to more comprehensive reporting standards, exemplified by the TRIPOD statement (Collins et al., 2015). More recent methodological work further indicates that the field is moving beyond model development toward questions of external validation, generalisability, usability, and implementation in real-world clinical settings (Carriero et al., 2026). Thus, the evolution of ML-based prediction is increasingly characterized not only by algorithmic innovation but also by greater emphasis on methodological robustness and translational value.

At the field level, the rapid growth and diversification of ML-based clinical prediction research have created a complex research landscape spanning algorithms, diseases, data sources, validation strategies, and performance measures. Li et al. (2026) recently provided a bibliometric assessment of ML-based clinical prediction models, identifying rapid publication growth since 2015 and prominent research themes in algorithm optimisation, multimodal data integration, and model interpretability. However, their analysis mainly characterises the global research structure and thematic hotspots of ML-based clinical prediction models, while the methodological evolution linking statistical learning, predictive performance, and model validation remains less explicitly synthesised.

RESEARCH GAP

Despite substantial methodological and application-oriented research, important gaps remain in understanding how ML-based disease prediction has evolved as a research field. Christodoulou et al. (2019) demonstrated that ML did not consistently outperform logistic regression and highlighted weaknesses in validation and calibration, while Binuya et al. (2022) emphasised the need to evaluate discrimination, calibration, clinical usefulness, and model updating. Andaur Navarro et al. (2021) further showed substantial deficiencies in reporting supervised ML prediction models, indicating persistent concerns about transparency and reproducibility. More recently, Collins et al. (2015) and Carriero et al. (2026) have emphasised standardised reporting, external validation, generalisability, and clinical implementation as increasingly important dimensions of prediction-model research.

Although Li et al. (2026) recently mapped the global research landscape of ML-based clinical prediction models using bibliometric methods, their analysis focused primarily on publication growth, collaboration patterns, intellectual structure, and thematic hotspots. Conse-

quently, researchers have insufficiently characterised the field's longitudinal evolution across the interconnected dimensions of statistical learning, model validation, and predictive performance. To address this gap, the present study uses bibliometric and science-mapping approaches to examine how these methodological dimensions have evolved, interacted, and shaped emerging research directions in ML-based disease prediction.

MATERIAL AND METHOD

Study Design and Reporting Framework

This study applied a bibliometric analysis and science mapping approach to quantitatively examine the global evolution of machine learning (ML)-based human disease prediction. Bibliometric methodology enables systematic investigation of publication trends, citation structures, intellectual development, collaboration networks, and emerging research themes within a scientific domain through quantitative analysis of published literature (Donthu et al., 2021).

The literature identification and selection process was reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA 2020) statement (Page et al., 2021), to ensure transparency and reproducibility. While PRISMA was originally developed for systematic reviews evaluating clinical interventions and outcomes, its structured identification, screening, eligibility, and inclusion framework has been adapted by bibliometric researchers to enhance the transparency of literature curation and dataset preparation. Accordingly, PRISMA was used in the present study as a reporting and screening framework for dataset preparation rather than as a tool for assessing risk of bias in individual studies.

Data Source and Search Strategy

The Dimensions AI database (Digital Science & Research Solutions Inc.) was used for the literature search owing to its broad coverage of scientific publications, citation data, author affiliations, and research output across biomedical sciences, computer science, and healthcare analytics (Thelwall, 2018). The database search was conducted and the dataset exported on 7 September 2026, covering publications from 2000 to July 2026. The search was performed on the title and abstract fields, with the goal of maximizing retrieval sensitivity while maintaining search specificity. The search strategy consisted of three conceptual components: machine learning and predictive modelling terms, disease prediction and clinical outcome terms, and human population terms. The complete search string is provided in Appendix A.

Eligibility Criteria

The records were included if they met all of the following criteria: reported the development, validation, and/or performance evaluation of a machine learning-based predictive model; addressed a human disease-related outcome (diagnosis, prognosis, risk stratification, or disease-related classification); were published as a peer-reviewed journal article, conference proceeding, or review indexed in Dimensions AI within the specified date range; and

were available in English either as full text or as a title/abstract sufficient for screening.

Studies were excluded if they focused on non-human subjects (animal, veterinary, or purely computational/simulated populations without clinical relevance); did not report a clearly defined predictive outcome; were duplicate records or records marked as ineligible for indexing; or could not be retrieved in full text where full-text assessment was required for eligibility confirmation.

Study Selection Process

Figure 1 illustrates the PRISMA-based study selection procedure adopted for this bibliometric analysis. A total of 3,152 records were identified from the Dimensions AI database. During the identification phase, 30 duplicate records and 2 records marked as ineligible were removed, resulting in 3,120 records for title and abstract screening. Following initial screening, 132 records were excluded due to irrelevance to the research objectives. Subsequently, 2,988 reports were sought for retrieval, of which 3 could not be retrieved. The remaining 2,985 studies underwent full-text eligibility assessment, at which stage 4 reports were excluded because they were not related to human disease prediction and 2 studies lacked a clearly defined predictive outcome. Finally, 2,979 studies met the predefined eligibility criteria and were included in the bibliometric analysis.

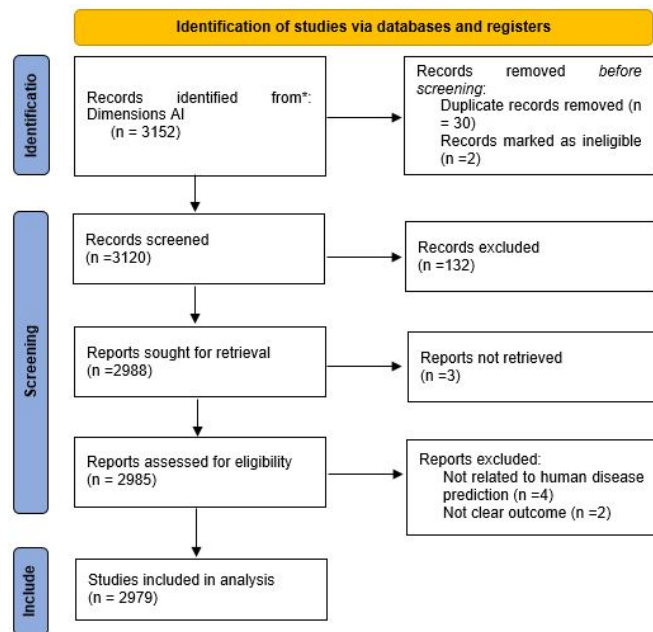


Figure 1. PRISMA 2020 framework adopted for this study.

Data Extraction, Dataset Preparation, and Metadata Quality Assessment

Bibliographic records were exported in comma-separated value (CSV) format containing metadata fields available in Dimensions AI relevant to bibliometric analysis, including author information (AU), cited references (CR), document type (DT), publication year (PY), publication title (TI), total citations (TC), source/journal information (SO), Digital Object Identifier (DI), abstract (AB), author affiliation (C1), and corresponding author

information (RP). The exported dataset was manually checked and cleaned prior to analysis. This process included deduplication, standardization of author name variants, and removal of residual records that did not meet the eligibility criteria.

Prior to conducting the bibliometric analyses, the completeness of each extracted metadata field was systematically assessed by calculating the number and percentage of missing records per field across the final dataset of 2,979 records. Fields with missingness below 1% were classified as having excellent-to-good completeness and were retained for direct analysis; fields with substantial or complete missingness were used cautiously.

Bibliometric Analysis and Science Mapping

Bibliometric analysis was performed to quantitatively describe the productivity, impact, collaborative structure, and conceptual development of research advancing human disease prediction via machine learning techniques. Analyses were organized into two complementary components: performance analysis, used to measure the productivity and impact of research entities, and science mapping, used to investigate relational and conceptual structures using network-based statistical methods (Donthu et al., 2021; Aria and Cuccurullo, 2017).

Publication Productivity and Growth Analysis

The temporal evolution of research output was assessed through annual publication productivity. The annual growth rate (AGR) of publications was calculated as

$$AGR_t = \frac{P_t - P_{t-1}}{P_{t-1}} \times 100, \quad (1)$$

where P_t represents the number of publications in year t and P_{t-1} represents the number of publications in the preceding year.

The compound annual growth rate (CAGR) over the study period was estimated as

$$CAGR = \left[\left(\frac{P_f}{P_i} \right)^{1/n} - 1 \right] \times 100, \quad (2)$$

where P_f and P_i represent the final and initial publication counts, respectively, and n denotes the number of years between observations. The reduced count for 2026 was interpreted in the context of incomplete indexing of the extraction year.

Citation Impact Analysis

Scientific influence was evaluated through citation-based indicators. The average citation impact (ACI) was calculated as

$$ACI_t = \frac{TC_t}{NP_t}, \quad (3)$$

where TC_t is the total number of citations received and NP_t is the total number of publications in year t .

Source and Institutional Productivity Analysis

The productivity of scientific sources (journals) and research affiliations was assessed by quantifying the cumulative number of publications contributed by each source and institution over time. Sources and institutions were ranked according to total publication output to identify the most productive contributors to the field, and cumulative growth trajectories were plotted to characterize the timing and rate of their contributions.

Collaboration Network Analysis

Co-authorship networks at the author and country levels were used to examine scientific collaboration structures. Each researcher or country was represented as a node, and co-authorship relationships were represented as edges. Let $a_{ip} = 1$ if entity i contributed to publication p , and $a_{ip} = 0$ otherwise. The collaboration strength between entities i and j was represented by

$$w_{ij} = \sum_{p=1}^N a_{ip}a_{jp}, \quad (4)$$

where w_{ij} is the number of publications jointly contributed by entities i and j . Network visualization was performed using weighted and normalized association-strength algorithms (van Eck and Waltman, 2010).

Keyword and Title-Term Frequency Analysis

As author keywords, indexed keywords, and science category classifications were unavailable in the exported Dimensions AI dataset, a title-based term frequency analysis was performed as a substitute to identify dominant research concepts, consistent with approaches adopted in prior bibliometric studies relying on Dimensions AI (Li et al., 2026). Terms were extracted from article titles following pre-processing, including removal of stop words and irrelevant terms and standardization of synonymous terms, such as merging “ML” and “machine learning”. Term frequencies were computed and visualized, and temporal term-frequency patterns were examined to characterize the evolution of terminology.

Thematic Mapping and Evolution Analysis

Thematic mapping was used to assess the evolution and maturity of research themes, following the strategic diagram approach of Cobo et al. (2011) and the co-word analysis framework of Callon et al. (1991). The two dimensions were centrality, representing the degree of interaction of a theme with other themes, and density, representing internal development and coherence.

The centrality measure was represented as

$$C_k = \sum_{h \neq k} e_{hk}, \quad (5)$$

where e_{hk} represents the strength of external connections between theme nodes. Density was represented as

$$D_k = \frac{\sum_{e_{hk} \in k} e_{hk}}{w_k}, \quad (6)$$

where e_{hk} represents internal relationships between title-derived terms within a theme and w_k is the number of terms in the theme.

Based on centrality and density, themes were classified into four categories: motor, basic, niche or specialised, and emerging or declining themes. Motor themes are highly developed and central. Basic themes have high centrality but low density. Niche themes have low centrality but high density. Emerging or declining themes have both low centrality and low density.

To examine changes in research priorities over time, the study period was divided into two sub-periods (2001–2024 and 2025–2026), and a thematic evolution map was constructed to trace continuity, splitting, merging, and disappearance of themes.

Network Clustering and Community Detection

Research entities and concepts were partitioned into clusters using community detection algorithms. Clusters were identified by maximizing modularity (van Eck and Waltman, 2010):

$$Q = \frac{1}{2m} \sum_{i,j} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j), \quad (7)$$

where m represents the total number of network edges, A_{ij} represents the observed connection between nodes i and j , k_i and k_j represent their degrees, and $\delta(c_i, c_j)$ is an indicator equal to 1 if the nodes belong to the same community and 0 otherwise.

Software and Visualization Tools

All bibliometric analyses were conducted using R and the Bibliometrix package (Aria and Cuccurullo, 2017), accessed through its interactive web interface, Biblioshiny. Biblioshiny was used for metadata quality assessment, publication growth analysis, citation impact analysis, source and institutional productivity analysis, title-term frequency analysis, and thematic mapping and evolution analysis.

VOSviewer was used for constructing and visualizing the author collaboration network and country collaboration network (van Eck and Waltman, 2010). Network visualizations were generated using weighted and normalized association strength algorithms.

The graphical outputs produced included the publication growth trend plot, citation impact trend plot, cumulative source and institutional productivity plots, country collaboration network, leading-sources bar chart, author productivity/citation plot, thematic evolution map, strategic thematic map, author collaboration network, title-term frequency chart, and temporal term-evolution plot.

RESULTS AND DISCUSSION

The records in this dataset were highly complete for basic bibliographic data, such as author information, cited

references, publication year, document type, publication title, and number of citations. Minor missingness was found in journal information (0.13%), DOI information (0.20%), and abstracts (0.27%). Significant missingness was found with respect to author affiliation (31.96%). The exported Dimensions AI dataset lacked corresponding author information, author keywords, indexed keywords, language, and scientific categories.

Table 1. Assessment of metadata availability and completeness for bibliometric analysis

Field	Description	Missing Records	Missing (%)	Status
AU	Author information	0	0.00	Excellent
CR	Cited references	0	0.00	Excellent
DT	Document type	0	0.00	Excellent
PY	Publication year	0	0.00	Excellent
TI	Publication title	0	0.00	Excellent
TC	Total citations	0	0.00	Excellent
SO	Source/journal information	4	0.13	Good
DI	Digital Object Identifier	6	0.20	Good
AB	Abstract	8	0.27	Good
C1	Author affiliation	952	31.96	Limited
RP	Corresponding author	2979	100.00	Completely missing
DE	Author keywords	2979	100.00	Completely missing
ID	Keywords Plus/Indexed keywords	2979	100.00	Completely missing
LA	Language information	2979	100.00	Completely missing
WC	Science categories	2979	100.00	Completely missing

Publication Growth and Citation Impact

Findings demonstrate a rapid increase in scientific output on machine learning-based human disease prediction, particularly after 2020.

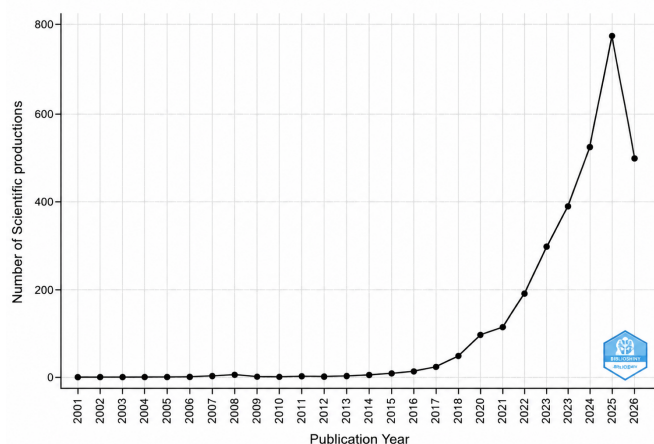


Figure 2. Temporal evolution of global scientific production in machine learning-based human disease prediction research from 2001 to 2026.

Publication growth accelerated sharply, reaching its highest level in 2025, indicating growing research interest and adoption of machine learning approaches in healthcare prediction models. The lower output in 2026 reflects an incomplete publication year due to data extraction timing (Figure 2).

Citation impact showed considerable fluctuations, with higher citation rates observed during the early years (2001–2003) and a notable peak in 2011. After a sharp decline in 2012, citation performance gradually recovered, reaching another elevated phase between 2017 and 2020. A declining trend was observed after 2020, likely reflecting the reduced citation accumulation period for recently published studies (Figure 3).

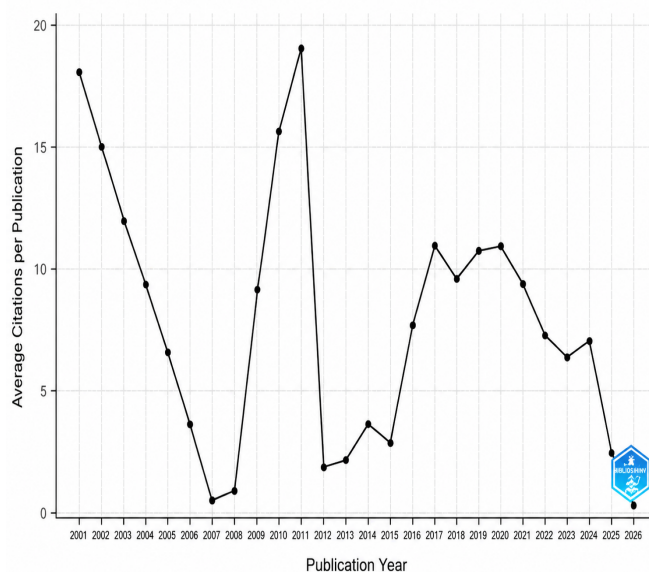


Figure 3. Temporal variation in average citation impact per publication in machine learning-based human disease prediction research.

Scientific Sources and Institutional Contributions

The publication output remained limited during the early years but increased substantially after 2018, reflecting accelerated research activity in this field. Scientific Reports demonstrated the highest cumulative growth, followed by the International Journal for Research in Applied Science and Engineering Technology and IEEE Access, indicating their prominent contribution to disseminating machine learning-based healthcare research.

Findings show that scientific contributions from leading institutions remained limited during the early period but increased substantially after 2018. University of Oxford demonstrated the earliest and most consistent growth, while Anna University, Chennai showed the highest recent publication output. Other institutions, including Harvard University, Princess Nourah Bint Abdulrahman University, and Vellore Institute of Technology University, exhibited rapid growth mainly after 2020.

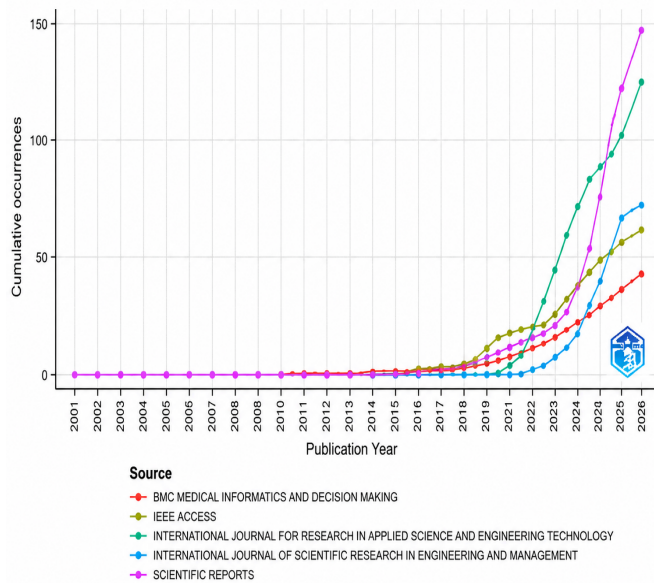


Figure 4. Cumulative publication growth of leading research sources in machine learning-based human disease prediction research.

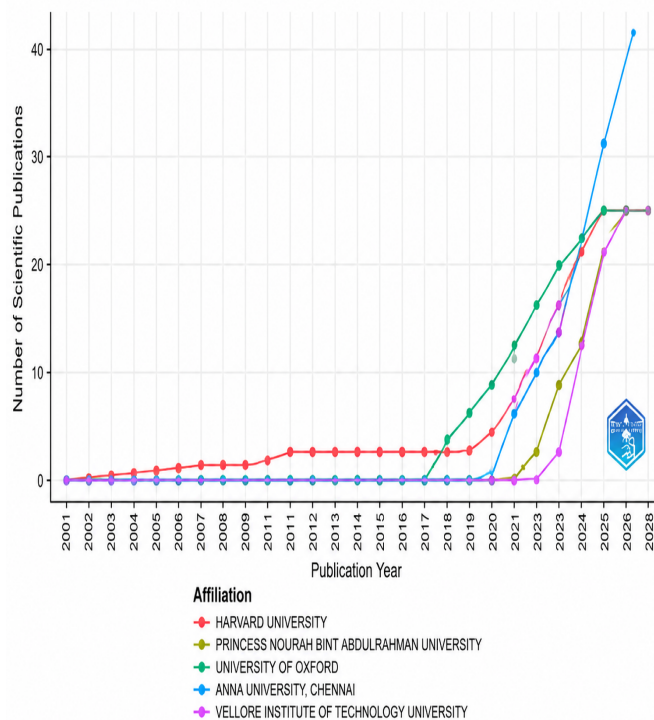


Figure 5. Temporal evolution of scientific contributions from leading research affiliations in machine learning-based human disease prediction research.

Collaboration and Intellectual Structure

The network demonstrates extensive global cooperation, with major research contributions originating from China, the United States, India, and the United Kingdom, which occupy central positions with high connectivity and collaboration intensity. The dense inter-country links indicate strong knowledge exchange and

cross-national research partnerships. The overlay visualization suggests that recent collaborative activities have increasingly concentrated around countries with active contributions during 2022–2025.

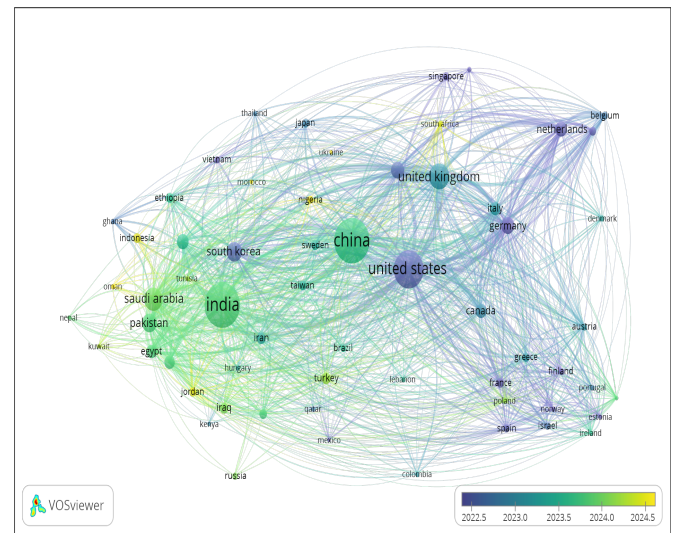


Figure 6. Global country collaboration network in machine learning-based human disease prediction research.

Scientific Reports emerged as the most productive source with the highest number of publications ($n = 148$), followed by the International Journal for Research in Applied Science and Engineering Technology ($n = 127$) and the International Journal of Scientific Research in Engineering and Management ($n = 74$). Biomedical and multidisciplinary journals, including IEEE Access, BMC Medical Informatics and Decision Making, and PLOS ONE, also demonstrated substantial contributions.

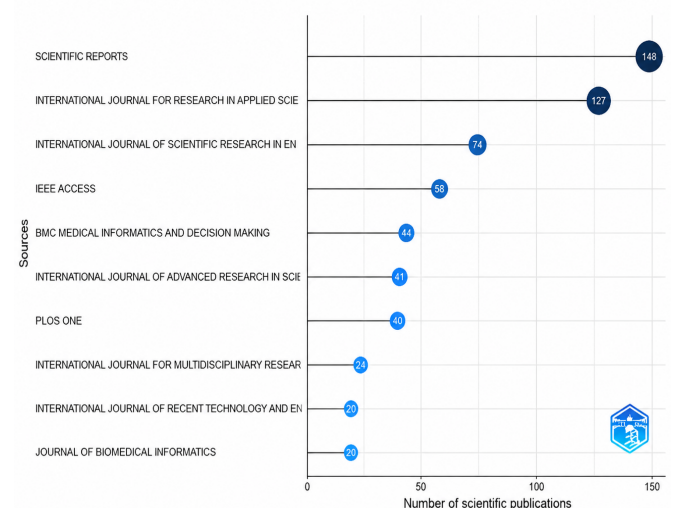


Figure 7. Most productive sources publishing research on machine learning-based human disease prediction.

The visualization indicates that leading contributors, including Wang Y, Zhang Y, Chen Y, Li Y, and Zhang X, have maintained sustained research activity primarily during the recent decade. Publication intensity increased markedly after 2020, with larger nodes representing years

of higher publication output and darker shades indicating greater citation impact.

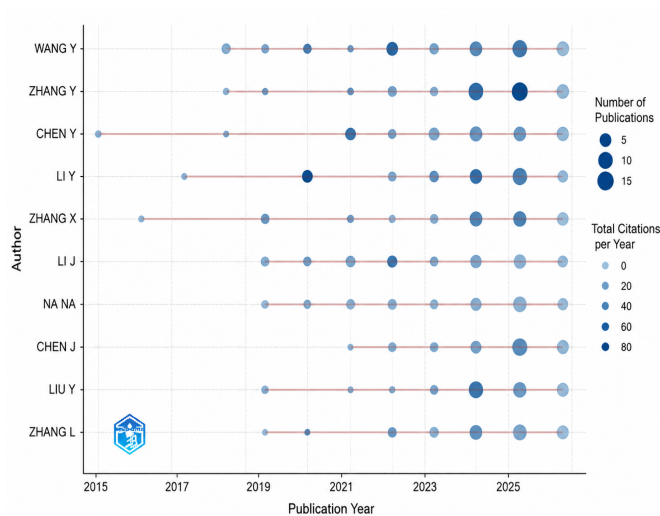


Figure 8. Temporal evolution and citation impact of leading authors in machine learning-based human disease prediction research.

Evolution of Research Themes

The earlier literature was primarily structured around the broad themes “prediction” and “learning.” In the recent period, these themes evolved into more application-oriented topics, particularly “disease,” “patients,” and “healthcare,” while “prediction” remained a persistent core theme. This transition suggests a shift from predominantly methodological machine-learning research toward more patient- and healthcare-focused disease prediction applications (Figure 9).

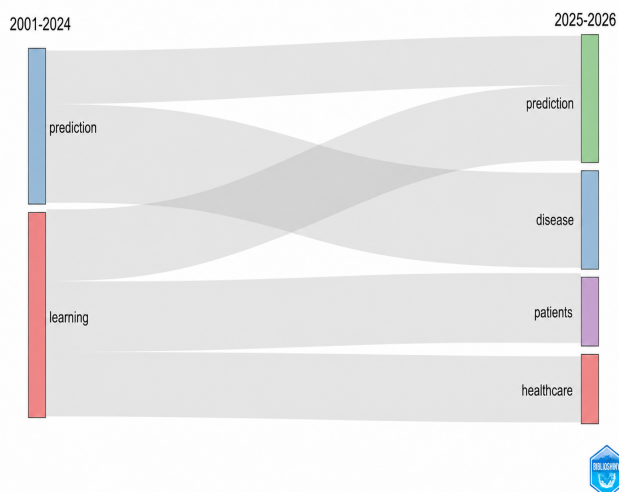


Figure 9. Thematic evolution of machine learning-based human disease prediction research across two time periods (2001–2024 and 2025–2026).

The quadrant analysis identifies “disease,” “heart,” and “deep” as well-developed and highly connected motor themes, indicating their strong influence and maturity within the research domain. The themes “prediction,”

“learning,” and “machine” appear as basic themes, reflecting their fundamental importance and broad applicability across studies. The themes “patients,” “validation,” and “development” represent niche themes. Meanwhile, “healthcare,” “artificial,” and “review” are positioned as emerging or declining themes (Figure 10).

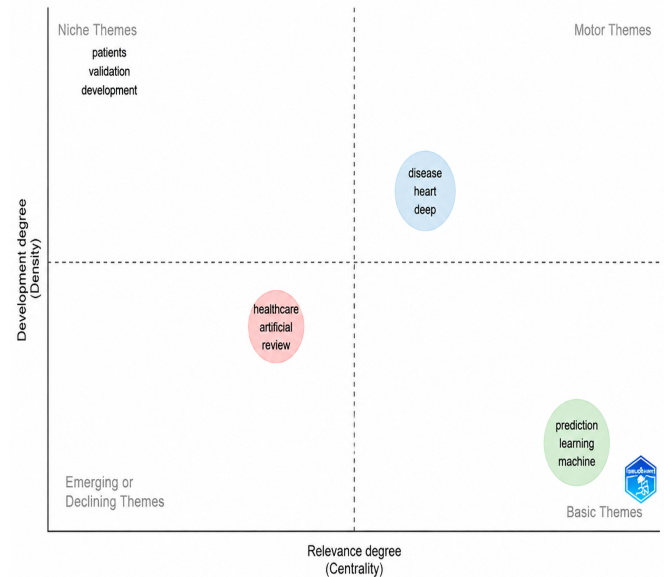


Figure 10. Strategic thematic map of machine learning-based human disease prediction research.

The network reveals a highly interconnected collaborative cluster dominated by authors such as Wang Y, Zhang Y, Chen Y, and Zheng J, who occupy central positions with extensive collaborative links. A smaller peripheral cluster involving authors such as Collins G, Riley RD, and Maj J indicates limited but distinct collaborative relationships.

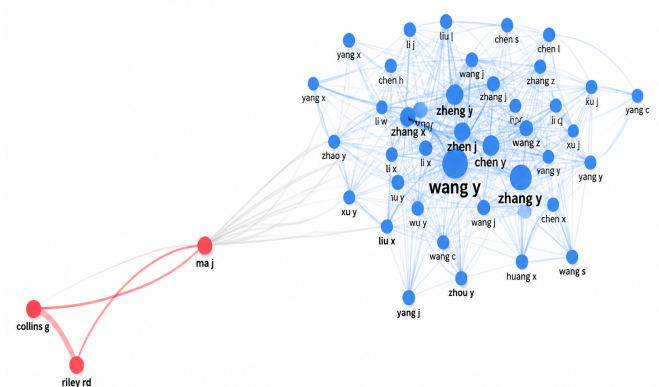


Figure 11. Author collaboration network in machine learning-based human disease prediction research.

Emerging Research Directions

The dominance of terms such as “prediction,” “disease,” “machine,” and “learning” reflects the central research focus on applying machine learning approaches for disease prediction. Other frequently occurring title terms,

including “model,” “clinical,” “risk,” “heart,” “diagnosis,” and “data,” indicate strong emphasis on predictive modelling, healthcare applications, and clinical decision-support systems. Emerging title concepts such as “deep learning,” “artificial intelligence,” and “neural networks” demonstrate the increasing integration of advanced computational techniques in disease prediction research.



Figure 12. Title-derived term frequency analysis revealing dominant research concepts in machine learning-based human disease prediction.

The analysis indicates a clear shift toward advanced machine learning and clinically oriented applications. Traditional computational terms such as “data mining,” “random forest,” and “neural network” appeared earlier, whereas recent years show increased prominence of “deep learning,” “artificial intelligence,” “clinical prediction,” “interpretable machine learning,” and “learning framework.” The increasing frequency of disease-focused terms such as “disease prediction,” “disease risk,” and “clinical decision” highlights the growing emphasis on translating machine learning methods into practical healthcare prediction systems.

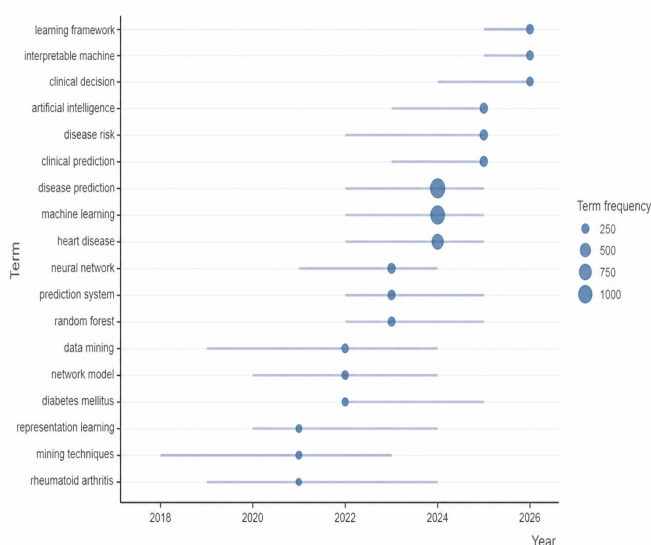


Figure 13. Temporal evolution of emerging research terms in machine learning-based human disease prediction.

DISCUSSION

This bibliometric analysis examines the evolution and intellectual structure of machine learning in human disease prediction, highlighting a shift towards clinically integrated, patient-centered systems. It notes a significant rise in scientific output after 2020, driven by the adoption of artificial intelligence, the availability of large healthcare datasets, and demand for predictive tools. This trend aligns with recognition of machine learning as a rapidly expanding field in biomedical informatics and precision medicine (Topol, 2019; Beam and Kohane, 2018).

The study indicates a rising trend in AI applications in healthcare, consistent with previous reviews. Key advances in computational power, EHRs, and algorithms have enabled machine learning in clinical prediction (Beam and Kohane, 2018). Rajkomar et al. (2019) further discussed the use of machine learning in medicine. The current research quantitatively confirms that disease prediction accelerated after 2020, coinciding with a surge in AI-driven healthcare studies.

Citation trajectories vary over time, with older studies generally having greater opportunity to accumulate citations than recent publications (Bornmann and Daniel, 2008; Ioannidis et al., 2014). Thus, the decline in average citations for new research does not necessarily signify a decrease in scientific importance; it can reflect a shorter citation window.

The analysis also indicates that multidisciplinary journals such as Scientific Reports, IEEE Access, and biomedical informatics journals contribute significantly to machine learning research in healthcare. This reflects the interdisciplinary nature of the field, which bridges computer science, statistics, engineering, and medicine (Dav-enport and Kalakota, 2019). Prior studies similarly emphasize the integration of algorithm development, validation methods, and healthcare expertise into clinical prediction (Esteva et al., 2019; Jiang et al., 2017).

Institutional and geographical analyses show global growth in healthcare research, particularly in AI. China, the United States, India, and the United Kingdom are key players with robust international collaboration. The analysis of author collaboration indicates highly interconnected research clusters led by contributors such as Wang Y, Zhang Y, Chen Y, and Zheng J. Such collaborative structures are important in fast-evolving scientific fields that require multidisciplinary expertise.

The thematic evolution analysis shows a shift from broad concepts such as machine learning to more specific themes related to disease and patient care. This change is consistent with the broader movement toward enhanced clinical decision-making and advanced pattern extraction from healthcare data (Topol, 2019; Miotto et al., 2018). Themes related to disease, heart health, and deep learning are prominent, while foundational themes include prediction, learning, and machine learning (LeCun et al., 2015). Cardiovascular disease prediction stands out as an important application, while deep learning has gained prominence for modelling complex relationships (Khera et al., 2021).

The growing attention to explainable artificial intel-

ligence (XAI) and interpretable machine learning addresses the limited interpretability of some existing models and the implications for clinical use (Holzinger et al., 2017; Rudin, 2019). The title-derived term analysis highlights the growing significance of deep learning, artificial intelligence, neural networks, clinical prediction, and disease risk modelling. Because the Dimensions AI dataset lacks author keywords, title-based analysis was used to identify conceptual trends.

However, challenges such as dataset bias, external validation, model generalisability, and clinical implementation persist. Current findings therefore suggest that future research should focus on prospective validation, transparency, fairness, and integration into clinical practice (Kelly et al., 2019; Nagendran et al., 2020).

The results broadly support the study's theoretical basis by showing the evolution of machine learning from conventional statistical prediction to more complex, data-driven, and clinically integrated predictive approaches. The growing prominence of deep learning, artificial intelligence, and sophisticated predictive models supports the proposition that ML can detect complex relationships and trends within extensive, diverse healthcare datasets. At the same time, the persistent prominence of "prediction" and "learning" demonstrates the continued relevance of statistical learning principles. The increasing focus on validation, interpretability, generalisability, and clinical deployment expands the theoretical basis beyond algorithmic performance. Thus, the results align with an evolutionary perspective progressing from algorithm development to predictive performance, validation, interpretability, and ultimately clinical implementation.

CONCLUSION

This bibliometric study analyses the evolution and future directions of machine learning in human disease prediction from 2001 to 2026. It highlights significant growth after 2020, driven by the integration of AI in healthcare. The research has shifted from an algorithm focus to clinical applications, emphasising risk assessment and decision support, with major contributions from countries such as China, the USA, India, and the UK. Key themes include deep learning, cardiovascular applications, and explainable AI, indicating a shift towards interpretable, clinically deployable systems. Challenges such as validation and integration into clinical workflows persist, necessitating a focus on trustworthy AI frameworks and real-world applicability. The study provides a roadmap for advancing AI-driven precision healthcare and is beneficial for researchers and policymakers.

ACKNOWLEDGMENTS

The authors would like to thank the individuals and institutions who contributed to this study.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

FUNDING

No specific funding was reported for this study.

SEARCH STRATEGY

```
("machine learning" OR "supervised learning" OR
  "statistical learning"
OR "deep learning" OR "neural network*" OR "
  random forest*"
OR "support vector machine*" OR "gradient
  boosting" OR XGBoost)
AND
("disease prediction" OR "disease risk
  prediction"
OR "disease diagnosis prediction"
OR "disease outcome prediction"
OR "health risk prediction"
OR "clinical prediction model*"
OR "clinical risk prediction")
AND
(human* OR patient*)
NOT
(veterinary OR animal* OR livestock OR cattle
  OR bovine
OR swine OR poultry OR canine OR equine
OR "plant disease*" OR crop*)
```

REFERENCES

- Pei-Yi Wu, Chia-Wen Cheng, Chandra D. Kaddi, Janani Venugopalan, Robert Hoffman, and May D. Wang. -omic and electronic health record big data analytics for precision medicine. *IEEE Transactions on Biomedical Engineering*, 64(2):263–273, 2016. doi:[10.1109/TBME.2016.2573285](https://doi.org/10.1109/TBME.2016.2573285).
- M. Z. I. Chowdhury and T. C. Turin. Precision health through prediction modelling: Factors to consider before implementing a prediction model in clinical practice. *Journal of Primary Health Care*, 12(1):3–9, 2020. doi:[10.1071/HC19087](https://doi.org/10.1071/HC19087).
- A. Saelmans, T. Seinen, V. Pera, A. F. Markus, E. Fridgeirsson, L. H. John, L. Schiphof-Godart, P. Rijnbeek, J. Reys, and R. Williams. Implementation and updating of clinical prediction models: A systematic review. *Mayo Clinic Proceedings: Digital Health*, 3(3): 100228, 2025. doi:[10.1016/j.mcpdig.2025.100228](https://doi.org/10.1016/j.mcpdig.2025.100228).
- S. K. Dhillon, M. D. Ganggayah, S. Sinnadurai, P. Lio, and N. A. Taib. Theory and practice of integrating machine learning and conventional statistics in medical data analysis. *Diagnostics*, 12(10):2526, 2022. doi:[10.3390/diagnostics12102526](https://doi.org/10.3390/diagnostics12102526).
- Ian A. Scott. Demystifying machine learning: A primer for physicians. *Internal Medicine Journal*, 51(9):1388–1400, 2021. doi:[10.1111/imj.15200](https://doi.org/10.1111/imj.15200).
- A. Al-Nafjan, A. Aljuhani, A. Alshebel, A. Alharbi, and A. Alshehri. Artificial intelligence in predictive healthcare: A systematic review. *Journal of Clinical Medicine*, 14(19):6752, 2025. doi:[10.3390/jcm14196752](https://doi.org/10.3390/jcm14196752).

- Kyle M. Kingsmore, C. E. Puglisi, A. C. Grammer, and Peter E. Lipsky. An introduction to machine learning and analysis of its use in rheumatic diseases. *Nature Reviews Rheumatology*, 17:710–730, 2021. doi:[10.1038/s41584-021-00708-w](https://doi.org/10.1038/s41584-021-00708-w).
- Gary S. Collins, Johannes B. Reitsma, Douglas G. Altman, and Karel G. M. Moons. Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (tripod): The tripod statement. *Annals of Internal Medicine*, 162(1):55, 2015. doi:[10.7326/M14-0697](https://doi.org/10.7326/M14-0697).
- C. L. Andaur Navarro, J. A. A. Damen, T. Takada, S. W. J. Nijman, P. Dhiman, J. Ma, G. S. Collins, R. Bajpai, R. D. Riley, K. G. M. Moons, and L. Hooft. Systematic review finds “spin” practices and poor reporting standards in studies on machine learning-based prediction models. *Journal of Clinical Epidemiology*, 158:99–110, 2023. doi:[10.1016/j.jclinepi.2023.03.024](https://doi.org/10.1016/j.jclinepi.2023.03.024).
- C. L. Andaur Navarro, J. A. A. Damen, T. Takada, S. W. J. Nijman, P. Dhiman, J. Ma, G. S. Collins, R. Bajpai, R. D. Riley, K. G. M. Moons, and L. Hooft. Risk of bias in studies on prediction models developed using supervised machine learning techniques: Systematic review. *BMJ*, 375:n2281, 2021. doi:[10.1136/bmj.n2281](https://doi.org/10.1136/bmj.n2281).
- Evangelia Christodoulou, Jun Ma, Gary S. Collins, Ewout W. Steyerberg, Jan Y. Verbakel, and Ben Van Calster. A systematic review shows no performance benefit of machine learning over logistic regression for clinical prediction models. *Journal of Clinical Epidemiology*, 110:12–22, 2019. doi:[10.1016/j.jclinepi.2019.02.004](https://doi.org/10.1016/j.jclinepi.2019.02.004).
- A. Zhang, L. Xing, J. Zou, and J. C. Wu. Shifting machine learning for healthcare from development to deployment and from models to data. *Nature Biomedical Engineering*, 6:1330–1345, 2022. doi:[10.1038/s41551-022-00898-y](https://doi.org/10.1038/s41551-022-00898-y).
- Q. Li, M. Zeng, D. Liang, X. Zheng, X. Zhang, and R. Martin-Payo. Research on machine learning-based clinical prediction models: A bibliometric analysis. *Frontiers in Oncology*, 16:1786176, 2026. doi:[10.3389/fonc.2026.1786176](https://doi.org/10.3389/fonc.2026.1786176).
- Naveen Donthu, Satish Kumar, Debmalya Mukherjee, Nitesh Pandey, and Weng Marc Lim. How to conduct a bibliometric analysis: An overview and guidelines. *Journal of Business Research*, 133:285–296, 2021. doi:[10.1016/j.jbusres.2021.04.070](https://doi.org/10.1016/j.jbusres.2021.04.070).
- Massimo Aria and Corrado Cuccurullo. bibliometrix: An r-tool for comprehensive science mapping analysis. *Journal of Informetrics*, 11(4):959–975, 2017. doi:[10.1016/j.joi.2017.08.007](https://doi.org/10.1016/j.joi.2017.08.007).
- Igor Kononenko. Machine learning for medical diagnosis: history, state of the art and perspective. *Artificial Intelligence in Medicine*, 23(1):89–109, 2001. doi:[10.1016/S0933-3657\(01\)00077-X](https://doi.org/10.1016/S0933-3657(01)00077-X).
- T. Fatima and A. Masood. Impact of digital leadership on open innovation: a moderating serial mediation model. *Journal of Knowledge Management*, 28(1):161–180, 2024. doi:[10.1108/JKM-11-2022-0872](https://doi.org/10.1108/JKM-11-2022-0872).
- R. Islam, A. Sultana, and M. R. Islam. A comprehensive review for chronic disease prediction using machine learning algorithms. *Journal of Electrical Systems and Information Technology*, 11(1):27, 2024. doi:[10.1186/s43067-024-00150-4](https://doi.org/10.1186/s43067-024-00150-4).
- R. Qasrawi, G. Issa, S. Thwib, R. AbuGhoush, M. Amro, R. Ayyad, and J. D. Kong. The role of machine learning in infectious disease early detection and prediction in the mena region: A systematic review. *Informatics in Medicine Unlocked*, 56:101651, 2025. doi:[10.1016/j.imu.2025.101651](https://doi.org/10.1016/j.imu.2025.101651).
- M. A. E. Binuya, E. G. Engelhardt, W. Schats, M. K. Schmidt, and E. W. Steyerberg. Methodological guidance for the evaluation and updating of clinical prediction models: a systematic review. *BMC Medical Research Methodology*, 22(1):316, 2022. doi:[10.1186/s12874-022-01801-8](https://doi.org/10.1186/s12874-022-01801-8).
- A. Carriero, A. A. de Hond, K. G. Moons, and M. van Smeden. Preliminary appraisal of machine learning based prediction models. *Journal of Clinical Epidemiology*, page 112256, 2026. doi:[10.1016/j.jclinepi.2026.112256](https://doi.org/10.1016/j.jclinepi.2026.112256).
- Matthew J. Page, Joanne E. McKenzie, Patrick M. Bossuyt, Isabelle Boutron, Tammy C. Hoffmann, Cynthia D. Mulrow, Larissa Shamseer, Jennifer M. Tetzlaff, Elie A. Akl, Sue E. Brennan, Roger Chou, Julie Glanville, Jeremy M. Grimshaw, Asbjorn Hrobjartsson, Manoj M. Lalu, Tianjing Li, Elizabeth W. Loder, Evan Mayo-Wilson, Steve McDonald, Luke A. McGuinness, Lesley A. Stewart, James Thomas, Andrea C. Tricco, Vivian A. Welch, Penny Whiting, and David Moher. The prisma 2020 statement: an updated guideline for reporting systematic reviews. *BMJ*, 372:n71, 2021. doi:[10.1136/bmj.n71](https://doi.org/10.1136/bmj.n71).
- Mike Thelwall. Dimensions: A competitor to scopus and the web of science? *Journal of Informetrics*, 12(2):430–435, 2018. doi:[10.1016/j.joi.2018.03.006](https://doi.org/10.1016/j.joi.2018.03.006).
- Nees Jan van Eck and Ludo Waltman. Software survey: Vosviewer, a computer program for bibliometric mapping. *Scientometrics*, 84(2):523–538, 2010. doi:[10.1007/s11192-009-0146-3](https://doi.org/10.1007/s11192-009-0146-3).
- Manuel J. Cobo, Antonio G. López-Herrera, Enrique Herrera-Viedma, and Francisco Herrera. An approach for detecting, quantifying, and visualizing the evolution of a research field: A practical application to the fuzzy sets theory field. *Journal of Informetrics*, 5(1):146–166, 2011. doi:[10.1016/j.joi.2010.10.002](https://doi.org/10.1016/j.joi.2010.10.002).
- Michel Callon, Jean-Pierre Courtial, and François Laville. Co-word analysis as a tool for describing the network of interactions between basic and technological research:

- The case of polymer chemistry. *Scientometrics*, 22(1): 155–205, 1991. doi:[10.1007/BF02019280](https://doi.org/10.1007/BF02019280).
- Eric J. Topol. High-performance medicine: The convergence of human and artificial intelligence. *Nature Medicine*, 25:44–56, 2019. doi:[10.1038/s41591-018-0300-7](https://doi.org/10.1038/s41591-018-0300-7).
- Andrew L. Beam and Isaac S. Kohane. Big data and machine learning in health care. *JAMA*, 319(13):1317–1318, 2018. doi:[10.1001/jama.2017.18391](https://doi.org/10.1001/jama.2017.18391).
- Alon Rajkomar, Jeffrey Dean, and Isaac Kohane. Machine learning in medicine. *New England Journal of Medicine*, 380:1347–1358, 2019. doi:[10.1056/NEJMr1814259](https://doi.org/10.1056/NEJMr1814259).
- Lutz Bornmann and Hans-Dieter Daniel. What do citation counts measure? a review of studies on citing behavior. *Journal of Documentation*, 64(1):45–80, 2008. doi:[10.1108/00220410810844150](https://doi.org/10.1108/00220410810844150).
- John P. A. Ioannidis, Kevin W. Boyack, Henry Small, Aaron A. Sorensen, and Richard Klavans. Bibliometrics: Is your most cited work your best? *Nature*, 514: 561–562, 2014. doi:[10.1038/514561a](https://doi.org/10.1038/514561a).
- Thomas Davenport and Ravi Kalakota. The potential for artificial intelligence in healthcare. *Future Healthcare Journal*, 6(2):94–98, 2019. doi:[10.7861/futurehosp.6-2-94](https://doi.org/10.7861/futurehosp.6-2-94).
- Andre Esteva, Alexandre Robicquet, Bharath Ramsundar, Volodymyr Kuleshov, Marc DePristo, Katherine Chou, Claire Cui, Greg Corrado, Sebastian Thrun, and Jeff Dean. A guide to deep learning in healthcare. *Nature Medicine*, 25(1):24–29, 2019. doi:[10.1038/s41591-018-0316-z](https://doi.org/10.1038/s41591-018-0316-z).
- Fei Jiang, Yong Jiang, Hui Zhi, Yi Dong, Hao Li, Shan-hong Ma, Yilong Wang, Qiang Dong, Haipeng Shen, and Yizhou Wang. Artificial intelligence in healthcare: Past, present and future. *Stroke and Vascular Neurology*, 2(4), 2017. doi:[10.1136/svn-2017-000101](https://doi.org/10.1136/svn-2017-000101).
- Riccardo Miotto, Fei Wang, Shuang Wang, Xin Jiang, and Joel T. Dudley. Deep learning for healthcare: Review and opportunities. *Briefings in Bioinformatics*, 19:1236–1246, 2018. doi:[10.1093/bib/bbx044](https://doi.org/10.1093/bib/bbx044).
- Yann LeCun, Yoshua Bengio, and Geoffrey Hinton. Deep learning. *Nature*, 521(7553):436–444, 2015. doi:[10.1038/nature14539](https://doi.org/10.1038/nature14539).
- Rohan Khera, Jonathan Haimovich, Nicholas C. Hurley, Robert McNamara, John A. Spertus, Nihar R. Desai, and Harlan M. Krumholz. Use of machine learning models to predict death after acute myocardial infarction. *JAMA Cardiology*, 6(6):633–641, 2021. doi:[10.1001/jamacardio.2021.0122](https://doi.org/10.1001/jamacardio.2021.0122).
- Andreas Holzinger, Chris Biemann, Constantinos S. Pattichis, and Douglas B. Kell. What do we need to build explainable ai systems for the medical domain? *arXiv*, 2017. doi:[10.48550/arXiv.1712.09923](https://doi.org/10.48550/arXiv.1712.09923).
- Cynthia Rudin. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence*, 1:206–215, 2019. doi:[10.1038/s42256-019-0048-x](https://doi.org/10.1038/s42256-019-0048-x).
- Christopher J. Kelly, Alan Karthikesalingam, Mustafa Suleyman, Greg Corrado, and Dominic King. Key challenges for delivering clinical impact with artificial intelligence. *BMC Medicine*, 17(1):195, 2019. doi:[10.1186/s12916-019-1426-2](https://doi.org/10.1186/s12916-019-1426-2).
- Myura Nagendran, Yie Chen, C. A. Lovejoy, A. C. Gordon, Matthieu Komorowski, Hugh Harvey, and Mahiben Maruthappu. Artificial intelligence versus clinicians: systematic review of design, reporting standards, and claims of deep learning studies. *BMJ*, 368: m689, 2020. doi:[10.1136/bmj.m689](https://doi.org/10.1136/bmj.m689).

Citation: Kafle, S., Kushwaha, S., Adhikari, B.P., & Subedi, P.(2026). Evolution of Machine Learning-Based Human-Disease Prediction Models: A Bibliometric Analysis of Statistical Learning, Model Validation, and Predictive Performance. *Nepal Journal of Statistics and Data Sciences*. 1(1),38–49. DOI: [10.68031/njsd.v1i.00005](https://doi.org/10.68031/njsd.v1i.00005)